THE NATIONAL UNIVERSITY OF LESOTHO
Department of Mathematics and Computer Science — Faculty of Science and Technology

M3503 — Ordinary Differential Equations

Unit 2: Higher-Order Linear Differential Equations

Structure, solution methods, and the mathematics of oscillation

Homogeneous and Non-Homogeneous Differential Equations

Scope of these notes

This covers the introduction (with the operator L set up explicitly), a full deep dive into homogeneous equations with five worked examples, and a full deep dive into non-homogeneous equations with five worked examples and the standard trial-form table. Reduction of order and the remaining solution methods follow separately later.

1 Introduction: The Operator L

Setting up notation once, properly

Everything in this unit is written in terms of one object, so it is worth building it carefully, once, right at the start, every later section just reuses this same L.

Write $D = \dfrac{d}{dx}$ for the derivative operator: $Dy = y'$, $D^2y = y''$, and in general $D^k y = y^{(k)}$. An n-th order linear ODE, after dividing through by its leading coefficient so the $y^{(n)}$ term has coefficient 1, can always be written as

$$y^{(n)} + p_{n-1}(x)y^{(n-1)} + \cdots + p_1(x)y' + p_0(x)y = g(x)$$

Using D, rewrite the left-hand side as

$$L := D^n + p_{n-1}(x)D^{n-1} + \cdots + p_1(x)D + p_0(x)$$

so that the whole equation becomes simply $L[y] = g(x)$.

L is nothing more than a sum of derivatives

Read the definition of L literally: it is a *sum*, term by term, of derivative operators D^k, each one scaled by a coefficient function $p_k(x)$ (with $p_n = 1$ after normalising). That is all L is — no more, no less:

$$L[y] = \underbrace{D^n y}_{y^{(n)}} + p_{n-1}(x)\underbrace{D^{n-1}y}_{y^{(n-1)}} + \cdots + p_1(x)\underbrace{Dy}_{y'} + p_0(x)\underbrace{y}_{D^0 y}$$

Every single equation in this unit is one of exactly two forms: $L[y] = 0$ (**homogeneous** — §2) or $L[y] = g(x)$ (**non-homogeneous** — §3). Once L is fixed for a given equation, both sections are just asking different questions about the *same* object. Keep this definition in mind; nothing new is introduced later — L is reused exactly as defined here, every time.

Why L is linear, directly from this definition

Each individual piece D^k is linear: $D^k(c_1y_1 + c_2y_2) = c_1D^ky_1 + c_2D^ky_2$, because differentiation distributes over addition and pulls out constants. Summing finitely many linear pieces (each $p_k(x)D^k$) is still linear. So directly from the sum-of-derivatives definition:

$$L[c_1y_1 + c_2y_2] = c_1L[y_1] + c_2L[y_2]$$

This one property is what makes the entire unit tractable: it is the reason superposition works for the homogeneous case (§2), and the reason the $y = y_c + y_p$ split works for the non-homogeneous case (§3). Both sections below trace back to this single fact about L.

2 Homogeneous Equations: $L[y] = 0$

The general homogeneous picture

Setting $g(x) = 0$ in $L[y] = g(x)$ gives $L[y] = 0$. Because L is linear (shown above), if $y_1, \dots, y_n$ each satisfy $L[y] = 0$, then so does *any* combination $c_1y_1 + \cdots + c_ny_n$. This is the superposition principle, and it holds for *any* coefficients $p_k(x)$, constant or not. If n such solutions are linearly independent (tested by the Wronskian), then $y_c = c_1y_1 + \cdots + c_ny_n$ is the general solution, there is nothing else out there to find.

The rest of this section restricts to the case that is actually solvable by hand directly: **constant coefficients**, where every $p_k(x)$ is a constant a_k, so

$$L = D^n + a_{n-1}D^{n-1} + \cdots + a_1D + a_0$$

Why $y = e^{rx}$ is the right guess

Try $y = e^{rx}$ inside L. Since $De^{rx} = re^{rx}$, applying D repeatedly just multiplies by r each time: $D^ke^{rx} = r^ke^{rx}$. Substitute this into L's definition — literally replace each D^k with r^k, exactly as L is built:

$$L[e^{rx}] = \big(r^n + a_{n-1}r^{n-1} + \cdots + a_1r + a_0\big)e^{rx} =: \phi(r)\, e^{rx}$$

The polynomial $\phi(r)$, obtained by literally swapping every D in L for r, is called the **characteristic polynomial**. Since e^{rx} is never zero, $y = e^{rx}$ solves $L[y] = 0$ exactly when $\phi(r) = 0$: the calculus problem $L[y] = 0$ has collapsed entirely into the algebra problem $\phi(r) = 0$.

The three second-order cases

For $L = D^2 + aD + b$ (so $\phi(r) = r^2 + ar + b$), the roots of $\phi(r) = 0$ fall into exactly three cases:

- **Distinct real roots** $r_1 \neq r_2$: two self-reproducing solutions come for free. $y_c = c_1e^{r_1x} + c_2e^{r_2x}$.
- **Repeated root** r: ϕ has a double root, so only one solution e^{rx} emerges directly — reduction of order (covered later) supplies the second, xe^{rx}. $y_c = (c_1 + c_2x)e^{rx}$.
- **Complex roots** $r = \alpha \pm i\beta$: Euler's formula repackages the complex exponentials as real functions. $y_c = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$.

For higher order, the same idea scales directly: factor $\phi(r)$ completely; a real root of multiplicity m contributes $(c_0 + c_1x + \cdots + c_{m-1}x^{m-1})e^{rx}$, and a complex conjugate pair of multiplicity m contributes the analogous cos / sin terms with polynomial coefficients.

Example H.1 — distinct real roots

$y'' - 5y' + 6y = 0$. Here $L = D^2 - 5D + 6$, so $\phi(r) = r^2 - 5r + 6$. Factor: $\phi(r) = (r - 2)(r - 3)$, roots $r = 2, 3$ — distinct and real.

$$y = c_1e^{2x} + c_2e^{3x}$$

Example H.2 — repeated root

$y'' - 4y' + 4y = 0$. Here $\phi(r) = r^2 - 4r + 4 = (r - 2)^2$, a double root at $r = 2$.

$$y = (c_1 + c_2x)e^{2x}$$

Example H.3 — complex roots

$y'' + 4y' + 13y = 0$. Here $\phi(r) = r^2 + 4r + 13$. The quadratic formula gives

$$r = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = -2 \pm 3i$$

so $\alpha = -2$, $\beta = 3$:

$$y = e^{-2x}(c_1 \cos 3x + c_2 \sin 3x)$$

Example H.4 — third order, a triple root

$y''' - 3y'' + 3y' - y = 0$. Here $L = D^3 - 3D^2 + 3D - 1$, so $\phi(r) = r^3 - 3r^2 + 3r - 1$. Notice the coefficients $1, -3, 3, -1$ are exactly the binomial expansion of $(r-1)^3$:

$$\phi(r) = (r-1)^3$$

a single root $r = 1$ with multiplicity 3. Following the multiplicity-m rule above ($m = 3$):

$$y = (c_1 + c_2 x + c_3 x^2)e^x$$

Example H.5 — fourth order, mixing real and complex roots

$y'''' - y = 0$. Here $L = D^4 - 1$, so $\phi(r) = r^4 - 1$. Factor completely, first as a difference of squares, then again:

$$\phi(r) = r^4 - 1 = (r^2 - 1)(r^2 + 1) = (r-1)(r+1)(r^2+1)$$

giving roots $r = 1$, $r = -1$ (distinct real), and $r = \pm i$ (complex, with $\alpha = 0, \beta = 1$). Reading off one contribution per root:

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$$

This is worth pausing on: a single fourth-order equation can package *both* of the real-root behaviours from §2 (H.1–H.2) *and* the oscillatory behaviour of H.3, all at once, once $\phi(r)$ is factored completely. Nothing new is needed beyond factoring the characteristic polynomial and reading off one contribution per distinct root (or root pair).

The physical heart of this section — vibrations and circuits

A mass on a spring with damping obeys $m\ddot{x} + c\dot{x} + kx = 0$ — literally $L[x] = 0$ for $L = mD^2 + cD + k$, relabelled with x in place of y and dots for time-derivatives. An RLC circuit obeys $L\ddot{q} + R\dot{q} + \frac{1}{C}q = 0$, the same equation again. The discriminant of $\phi(r)$ selects the physical regime directly out of the three-case table above:

- **Distinct real roots ↔ overdamped**: the system creeps back to rest, no oscillation — like a door closing through honey.
- **Repeated root ↔ critically damped**: the fastest return to rest with no overshoot — the design target for shock absorbers and door closers.
- **Complex roots ↔ underdamped**: decaying oscillation — the ring of a struck bell.

Because each complex root $r = \alpha \pm i\beta$ contributes $e^{\alpha x}(\cdots)$, the whole system is stable (settles to equilibrium) exactly when *every* root of $\phi(r)$ has negative real part — one criterion, read directly off the same polynomial used to build every example above.

3 Non-Homogeneous Equations: $L[y] = g(x)$

The $y = y_c + y_p$ split, from L's linearity

Now $g(x) \neq 0$: something is forcing the system. Because L is linear, if y_p is any *one* solution of $L[y] = g$, and y_c solves the homogeneous equation $L[y] = 0$ from §2, then

$$L[y_c + y_p] = L[y_c] + L[y_p] = 0 + g = g$$

so $y_c + y_p$ solves the forced equation. It is also the *complete* answer: if $\tilde{y}$ is any other solution of $L[y] = g$, then $L[\tilde{y} - y_p] = g - g = 0$, so $\tilde{y} - y_p$ is itself a homogeneous solution, meaning $\tilde{y} = y_c + y_p$ for some choice of constants. So the recipe is always: find y_c from §2 first, then find just *one* y_p.

Undetermined coefficients: why guessing y_p actually works

This section builds y_p for the case that comes up constantly in practice: constant coefficients, and $g(x)$ built from polynomials, exponentials, sines, and cosines (or products of these). These functions share a special property, differentiating any of them never produces a genuinely new kind of function; it stays within the same closed family. So a particular solution y_p, built from that same family with undetermined constant coefficients, is a legitimate guess: the equation cannot force y_p outside the family you started with.

If $g(x)$ contains…	**try y_p containing…**
a constant k	A
a polynomial $P_n(x)$ of degree n	$A_n x^n + \cdots + A_1 x + A_0$
e^{ax}	Ae^{ax}
$\cos \beta x$ or $\sin \beta x$	$A \cos \beta x + B \sin \beta x$
$e^{ax} P_n(x)$	$e^{ax}(A_n x^n + \cdots + A_0)$
$e^{ax} \cos \beta x$ or $e^{ax} \sin \beta x$	$e^{ax}(A \cos \beta x + B \sin \beta x)$

The modification rule

Suppose a term of your trial y_p already appears in y_c — say $g(x) = e^{ax}$ but a is a root of $\phi(r)$, the characteristic polynomial from §2. Then $L[Ae^{ax}] = A\phi(a)e^{ax} = 0$ (since $\phi(a) = 0$), so the naive trial can never equal a nonzero $g(x)$ — you would be trying to solve $0 = e^{ax}$. Multiplying the trial by x^s, where s is the multiplicity of a as a root of $\phi(r)$, breaks the degeneracy. This is the same idea as the "extra x" in §2's repeated-root case — here it is y_p, rather than a homogeneous solution, that needs the extra factor.

Recipe: (1) find y_c first (§2); (2) choose the trial from the table above; (3) if any term already appears in y_c, multiply that whole trial term by x^s; (4) substitute into $L[y] = g$, match coefficients, solve.

Example NH.1 — polynomial forcing, no overlap

$y'' - 3y' + 2y = 4x^2$. First, y_c: $\phi(r) = r^2 - 3r + 2 = (r-1)(r-2)$, so $y_c = c_1 e^x + c_2 e^{2x}$. Since $4x^2$ is a plain polynomial and no power of x overlaps with y_c's exponentials, try $y_p = Ax^2 + Bx + C$ directly. Then $y_p' = 2Ax + B$, $y_p'' = 2A$. Substitute:

$$2A - 3(2Ax + B) + 2(Ax^2 + Bx + C) = 4x^2$$

Matching powers of x: x^2: $2A = 4 \Rightarrow A = 2$. x^1: $-6A + 2B = 0 \Rightarrow B = 6$. x^0: $2A - 3B + 2C = 0 \Rightarrow C = 7$.

$$y = c_1 e^x + c_2 e^{2x} + 2x^2 + 6x + 7$$

Example NH.2 — exponential forcing, no overlap

$y''-3y'+2y=e^{3x}$. Same $y_c=c_1e^x+c_2e^{2x}$ as above. Here $a=3$ is *not* a root of $\phi(r)=(r-1)(r-2)$, so no modification is needed: try $y_p=Ae^{3x}$. Then $y_p'=3Ae^{3x}$, $y_p''=9Ae^{3x}$. Substitute:

$$9Ae^{3x}-3(3Ae^{3x})+2(Ae^{3x})=e^{3x} \implies (9-9+2)A=1 \implies A=\frac{1}{2}$$

$$y_p=\frac{e^{3x}}{2}$$

Example NH.3 — exponential forcing, single modification

$y''-3y'+2y=e^x$. Same y_c again — but now $g(x)=e^x$ *is already* one of the homogeneous solutions, since $r=1$ is a (simple) root of $\phi(r)$. The naive trial Ae^x would fail (it solves $L[y]=0$, not $L[y]=e^x$), so apply the modification rule with $s=1$: try $y_p=Axe^x$. A short product-rule computation for y_p', y_p'' and substitution collapses to $-Ae^x=e^x$, giving $A=-1$:

$$y_p=-xe^x$$

Example NH.4 — sinusoidal forcing, no overlap

$y''+y=\cos 2x$. Here $\phi(r)=r^2+1$, roots $r=\pm i$, so $y_c=c_1\cos x+c_2\sin x$ — the system's own natural frequency is 1. The forcing frequency is $2\neq 1$, so there is no overlap: try $y_p=A\cos 2x+B\sin 2x$. Then $y_p''=-4A\cos 2x-4B\sin 2x$. Substitute into $y''+y=\cos 2x$:

$$(-4A+A)\cos 2x+(-4B+B)\sin 2x=\cos 2x$$

Matching coefficients: $-3A=1\Rightarrow A=-\frac{1}{3}$; $-3B=0\Rightarrow B=0$.

$$y_p=-\frac{1}{3}\cos 2x$$

Example NH.5 — exponential forcing, double modification

$y''-2y'+y=e^x$. Here $\phi(r)=r^2-2r+1=(r-1)^2$: $r=1$ (of multiplicity 2) is a *repeated* root, so $y_c=(c_1+c_2x)e^x$. Now $g(x)=e^x$ overlaps y_c *twice over*: the naive trial Ae^x fails (it is in y_c), and even the singly-modified Axe^x fails (it is *also* in y_c, as the second homogeneous solution). With multiplicity $s=2$, try $y_p=Ax^2e^x$. Computing $y_p'=A(2x+x^2)e^x$, $y_p''=A(2+4x+x^2)e^x$ and substituting:

$$A(2+4x+x^2)e^x-2A(2x+x^2)e^x+Ax^2e^x=e^x$$

The bracket collapses: $(2+4x+x^2)-(4x+2x^2)+x^2=2$, so $2Ae^x=e^x\Rightarrow A=\frac{1}{2}$:

$$y_p=\frac{x^2e^x}{2}$$

Compare this directly against NH.3: same forcing e^x, but a repeated characteristic root here (versus a simple root there) doubles the power of x needed to escape y_c — $s=1$ there, $s=2$ here. This is exactly the general rule stated above, seen at both its simplest and its next level.

Where this shows up

Undetermined coefficients is the everyday tool for forced oscillations with clean inputs: a sinusoidal driving voltage $E_0\cos\omega t$ on a circuit (compare Example NH.4), a periodic push on a spring, a steady load on a beam. Matching the amplitude of y_p to the forcing frequency — and watching for the overlap that Examples NH.3 and NH.5 illustrate — is exactly how engineers compute a system's steady-state response and spot resonance: overlap between the forcing and the system's natural frequency is precisely what triggers the modification rule, and precisely

what causes real systems to resonate.

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Variation of Parameters

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Now let us consider the second-order ODE of the form

$$\boxed{y'' + P(x)y' + Q(x)y = f(x)} \tag{$*$}$$

We have seen under Undetermined Coefficients that the complementary solution y_c takes the form

$$\boxed{y_c = c_1 y_1 + c_2 y_2} \tag{1}$$

Analogous to this, we now ask whether the parameters c_1 and c_2 in y_c could be replaced with functions $u_1(x)$ and $u_2(x)$ respectively, so that

$$\boxed{y_p(x) = u_1 y_1 + u_2 y_2} \tag{2}$$

is a particular solution.

In order to solve for u_1 and u_2, we substitute derivatives of y_p from (2) into $(*)$, bearing in mind that y_c is the solution of the homogeneous part of $(*)$, i.e.

$$\boxed{y_p'' + Py_p' + Qy_p = 0}.$$

The substitution forces imposing conditions:

$$\left.\begin{array}{ll} ① & y_1u_1' + y_2u_2' = 0, \\ ② & y_1'u_1' + y_2'u_2' = f(x). \end{array}\right\} \text{ system} \tag{3}$$

These two equations help us solve for u_1 and u_2 using Cramer's rule.

System (3) is a linear system for u_1' and u_2' whose determinant is the Wronskian

$$W = W[y_1, y_2] = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1',$$

which is non-zero because $\{y_1, y_2\}$ is a fundamental set.

Cramer's rule gives the following from (3),

$$u_1' = -\frac{y_2 f}{W} \qquad \text{and} \qquad u_2' = \frac{y_1 f}{W}$$

$$\Rightarrow \quad u_1(x) = -\int \frac{y_2(x)f(x)}{W(x)}\,dx \qquad \text{and} \qquad u_2(x) = \int \frac{y_1(x)f(x)}{W(x)}\,dx.$$

Therefore

$$y_p(x) = -y_1(x)\int \frac{y_2(x)f(x)}{W(x)}\,dx + y_2(x)\int \frac{y_1(x)f(x)}{W(x)}\,dx.$$

Thus, we finally have the solution as

$$y(x) = y_c(x) + y_p(x).$$

Variation of parameters helps us solve ODEs which have $f(x)$ having the forms that are not included under the undetermined coefficients method (i.e. $f(x) = \ln(x)$, $\tan(x)$, $\frac{1}{x}$, etc).

Example

Solve

$$y'' + y' - 6y = 5e^{-3x}.$$

$$r^2 + r - 6 = 0 \;\Rightarrow\; r = -3,\; r = 2$$

$$\Rightarrow \quad y_c = c_1 e^{2x} + c_2 e^{-3x} \qquad \boxed{y_1 = e^{2x}, \quad y_2 = e^{-3x}}$$

Thus, $y_p = u_1 e^{2x} + u_2 e^{-3x}$.

$$W\left(e^{2x}, e^{-3x}\right) = \begin{vmatrix} e^{2x} & e^{-3x} \\ 2e^{2x} & -3e^{-3x} \end{vmatrix} = -3e^{-x} - 2e^{-x} = -5e^{-x}.$$

So

$$u_1 = -\int \frac{y_2 f}{W}\,dx = +\int \frac{e^{-3x}\cdot 5e^{-3x}}{5e^{-x}}\,dx = \int \frac{e^{-6x}}{e^{-x}}\,dx = \int e^{-5x}\,dx = -\frac{1}{5}e^{-5x} + c_1,$$

$$u_2 = \int \frac{y_1 f}{W}\,dx = \int \frac{e^{2x}\cdot 5e^{-3x}}{-5e^{-x}}\,dx = -\int dx = -x + c_2.$$

Therefore

$$y_p = -\frac{1}{5}e^{-5x}\cdot e^{2x} + (-x)e^{-3x} = -\frac{1}{5}e^{-3x} - xe^{-3x}.$$

Hence

$$y = c_1 e^{2x} + c_2 e^{-3x} - \frac{1}{5}e^{-3x} - xe^{-3x}.$$

** Verify the answer using undetermined coefficients method! **

Example

Solve

$$y'' + y = \sec x, \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Solution. $r^2 + 1 = 0 \Rightarrow y_1 = \cos x,\ y_2 = \sin x$, and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1.$$

$$u_1 = \int \frac{-\sin x\,\sec x}{1}\,dx = \int -\tan x\,dx = \ln|\cos x|,$$

$$u_2 = \int \frac{\cos x\,\sec x}{1}\,dx = \int dx = x.$$

Hence

$$y = c_1 \cos x + c_2 \sin x + \cos x\,\ln|\cos x| + x\sin x.$$

The Inverse Operator Method

1. The central idea: make y the subject

Consider a linear ODE with *constant* coefficients,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = f(x). \tag{1}$$

Its general solution is $y = y_c + y_p$, where the *complementary function* y_c solves the associated homogeneous equation (from the roots of the auxiliary equation, as in the previous methods) and y_p is any *particular integral*. This handout is about producing y_p quickly and systematically.

Write the differential operator

$$\mathrm{D} \equiv \frac{\mathrm{d}}{\mathrm{d}x}, \qquad \mathrm{D}^2 \equiv \frac{\mathrm{d}^2}{\mathrm{d}x^2}, \qquad \ldots$$

so that (1) collapses to $\phi(\mathrm{D})\, y = f(x)$, where $\phi(\mathrm{D}) = a_n \mathrm{D}^n + \cdots + a_1 \mathrm{D} + a_0$ is a *polynomial in the operator* D.

The idea is disarmingly simple. To solve the algebraic equation $5y = 10$ we divide both sides by 5 to **make y the subject**: $y = \frac{1}{5} \cdot 10$. We do exactly the same to $\phi(\mathrm{D})\, y = f(x)$ — divide both sides by the whole operator $\phi(\mathrm{D})$ to make y the subject:

Particular integral

$$y_p \;=\; \frac{1}{\phi(\mathrm{D})}\, f(x).$$

The symbol $\dfrac{1}{\phi(\mathrm{D})}$ is the **inverse operator** — by definition, the operator that undoes $\phi(\mathrm{D})$. Dividing by the operator is a *formal* (symbolic) division, but it obeys the ordinary rules of algebra, and that is precisely what makes the method work.

Everything that follows answers one question: *once y is the subject, how do we actually apply $\dfrac{1}{\phi(\mathrm{D})}$ to $f(x)$?* The technique depends entirely on what sits on the right-hand side, so we organise the method into **cases according to the form of $f(x)$**.

2. Solving by cases: what is on the right-hand side?

After making y the subject, identify $f(x)$ and jump to the matching case:

1. $f(x) = \mathrm{e}^{ax}$ — an exponential (replace $\mathrm{D} \to a$)
2. $f(x) = \sin ax$ or $\cos ax$ — a sinusoid (replace $\mathrm{D}^2 \to -a^2$)
3. $f(x) = x^m$ — a polynomial (expand $[\phi(\mathrm{D})]^{-1}$ in powers of D)
4. $f(x) = \mathrm{e}^{ax} V(x)$ — exponential times something (shift theorem)
5. $f(x) = x\, V(x)$ (product rule for operators)
6. anything else (partial fractions in D)

If $f(x)$ is a *sum* of these types, split it: apply $\dfrac{1}{\phi(\mathrm{D})}$ to each piece separately and add the results.

Case 1 — Right-hand side is an exponential, $f(x) = \mathrm{e}^{ax}$

Since $\mathrm{D}^n \mathrm{e}^{ax} = a^n \mathrm{e}^{ax}$, we have $\phi(\mathrm{D})\mathrm{e}^{ax} = \phi(a)\mathrm{e}^{ax}$, so applying the inverse operator simply *replaces* D *by* a:

Case 1

$$\frac{1}{\phi(\mathrm{D})}\,\mathrm{e}^{ax} = \frac{1}{\phi(a)}\,\mathrm{e}^{ax}, \qquad \phi(a) \neq 0.$$

Failure case ($\phi(a) = 0$, i.e. a is a root): use $\dfrac{1}{\phi(\mathrm{D})}\mathrm{e}^{ax} = \dfrac{x}{\phi'(a)}\mathrm{e}^{ax}$.

Example 1a — ordinary case

Solve $(\mathrm{D}^2 - 3\mathrm{D} + 2)y = \mathrm{e}^{3x}$.

Here $\phi(a) = a^2 - 3a + 2$, so $\phi(3) = 9 - 9 + 2 = 2 \neq 0$ and $y_\mathrm{p} = \frac{1}{2}\mathrm{e}^{3x}$. With $y_\mathrm{c} = c_1\mathrm{e}^{x} + c_2\mathrm{e}^{2x}$ (roots $a = 1, 2$), the general solution is

$$y = c_1\mathrm{e}^{x} + c_2\mathrm{e}^{2x} + \tfrac{1}{2}\mathrm{e}^{3x}.$$

Example 1b — resonant (failure) case

Solve $(\mathrm{D}^2 - 3\mathrm{D} + 2)y = \mathrm{e}^{2x}$.

Now $\phi(2) = 0$, so $a = 2$ is a simple root. Using $\phi'(\mathrm{D}) = 2\mathrm{D} - 3$ gives $\phi'(2) = 1$, hence $y_\mathrm{p} = \dfrac{x}{\phi'(2)}\mathrm{e}^{2x} = x\,\mathrm{e}^{2x}$.

Example 1c — lecture problem: a sum of exponentials

Solve completely $5y'' + 2y' - 24y = \mathrm{e}^{-2x} + 3\mathrm{e}^{x}$.

Operator form: $(5\mathrm{D}^2 + 2\mathrm{D} - 24)y = \mathrm{e}^{-2x} + 3\mathrm{e}^{x}$.

Complementary function. Auxiliary equation $5\mathrm{D}^2 + 2\mathrm{D} - 24 = 0$ has discriminant $2^2 + 4(5)(24) = 484 = 22^2$, so

$$\mathrm{D} = \frac{-2 \pm 22}{10} = 2 \quad \text{or} \quad -\tfrac{12}{5} \qquad \Rightarrow \qquad y_\mathrm{c} = C_1\mathrm{e}^{2x} + C_2\mathrm{e}^{-12x/5}.$$

Particular integral. Make y the subject and split the sum:

$$y_\mathrm{p} = \frac{1}{5\mathrm{D}^2 + 2\mathrm{D} - 24}\mathrm{e}^{-2x} + \frac{3}{5\mathrm{D}^2 + 2\mathrm{D} - 24}\mathrm{e}^{x}.$$

Replace $\mathrm{D} \to a$ in each. For $a = -2$: $5(4) + 2(-2) - 24 = -8$. For $a = 1$: $5 + 2 - 24 = -17$. Thus

$$y_\mathrm{p} = \frac{1}{-8}\mathrm{e}^{-2x} + \frac{3}{-17}\mathrm{e}^{x} = -\frac{1}{8}\mathrm{e}^{-2x} - \frac{3}{17}\mathrm{e}^{x},$$

and the general solution is

$$\boxed{y = C_1\mathrm{e}^{2x} + C_2\mathrm{e}^{-12x/5} - \tfrac{1}{8}\mathrm{e}^{-2x} - \tfrac{3}{17}\mathrm{e}^{x}.}$$

Watch the resonance condition. In Example 1c neither $a = -2$ nor $a = 1$ is a root of the auxiliary equation ($\phi(-2) = -8$ and $\phi(1) = -17$, both nonzero), so *there is no factor of* x in y_p. A common reflex is to attach an x to an exponential particular integral; the $\dfrac{x^r}{r!}$ form of

Case 1 applies *only* when $\phi(a) = 0$. Always test $\phi(a)$ before deciding.

Case 2 — Right-hand side is a sinusoid, $f(x) = \sin ax$ or $\cos ax$

Because $\mathrm{D}^2 \sin ax = -a^2 \sin ax$ (and likewise for cos), wherever ϕ can be expressed through D^2 we *replace* $\mathrm{D}^2 \to -a^2$. Any leftover odd power of D is cleared by multiplying above and below by a conjugate operator.

Case 2

$$\frac{1}{\phi(\mathrm{D}^2)} \sin ax = \frac{1}{\phi(-a^2)} \sin ax, \qquad \phi(-a^2) \neq 0.$$

Failure case ($\phi(-a^2) = 0$): use the standard resonant results

$$\frac{1}{\mathrm{D}^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax, \qquad \frac{1}{\mathrm{D}^2 + a^2} \cos ax = \frac{x}{2a} \sin ax.$$

Example 2a — ordinary case

Solve $(\mathrm{D}^2 + \mathrm{D} + 1)y = \sin 2x$.

Replace $D^2 = -a^2 \implies \mathrm{D}^2 \to -4$: $y_\mathrm{p} = \dfrac{1}{\mathrm{D} - 3} \sin 2x$. Rationalize with the conjugate $\mathrm{D} + 3$:

$$y_\mathrm{p} = \frac{\mathrm{D} + 3}{\mathrm{D}^2 - 9} \sin 2x = \frac{\mathrm{D} + 3}{-4 - 9} \sin 2x = -\frac{1}{13}\big(2 \cos 2x + 3 \sin 2x\big),$$

using $\mathrm{D} \sin 2x = 2 \cos 2x$ and $\mathrm{D}^2 \to -4$ once more in the denominator.

Example 2b — resonant (failure) case

Solve $(\mathrm{D}^2 + 4)y = \sin 2x$.

Here $\phi(-a^2) = -4 + 4 = 0$ with $a = 2$, so $y_\mathrm{p} = -\dfrac{x}{2 \cdot 2} \cos 2x = -\dfrac{x}{4} \cos 2x$.

Example 2c — lecture problem: clearing the leftover D

Solve completely $y'' - y' = \sin 2x$.

Operator form: $(\mathrm{D}^2 - \mathrm{D})y = \sin 2x$.

Complementary function. $\mathrm{D}^2 - \mathrm{D} = \mathrm{D}(\mathrm{D} - 1) = 0$ gives roots $0, 1$, so $y_\mathrm{c} = C_1 + C_2 \mathrm{e}^x$.

Particular integral. Make y the subject: $y_\mathrm{p} = \dfrac{1}{\mathrm{D}^2 - \mathrm{D}} \sin 2x$. Replace $\mathrm{D}^2 \to -4$:

$$y_\mathrm{p} = \frac{1}{-4 - \mathrm{D}} \sin 2x = -\frac{1}{\mathrm{D} + 4} \sin 2x.$$

A bare D remains, so multiply above and below by the conjugate $\mathrm{D} - 4$:

$$y_\mathrm{p} = -\frac{\mathrm{D} - 4}{\mathrm{D}^2 - 16} \sin 2x = -\frac{\mathrm{D} - 4}{-4 - 16} \sin 2x = \frac{1}{20}(\mathrm{D} - 4) \sin 2x.$$

Now $\mathrm{D} \sin 2x = 2 \cos 2x$, so

$$y_\mathrm{p} = \frac{1}{20}\big(2 \cos 2x - 4 \sin 2x\big) = \frac{1}{10}\big(\cos 2x - 2 \sin 2x\big),$$

and the general solution is

$$\boxed{y = C_1 + C_2 \mathrm{e}^x + \tfrac{1}{10}\big(\cos 2x - 2 \sin 2x\big).}$$

Check: substituting y_{p} gives $y_{\mathrm{p}}'' - y_{\mathrm{p}}' = \sin 2x$. ✓

Case 3 — Right-hand side is a polynomial, $f(x) = x^m$

Expand $\left[\phi(\mathrm{D})\right]^{-1}$ as a power series in D (a Maclaurin / geometric expansion), keeping terms only up to D^m, since $\mathrm{D}^{m+1}x^m = 0$.

Case 3

Factor out the constant term a_0 of $\phi(\mathrm{D})$, write the remainder as $1 \pm u$ with $u = O(\mathrm{D})$, and use

$$\frac{1}{1+u} = 1 - u + u^2 - \dots, \qquad \frac{1}{1-u} = 1 + u + u^2 + \dots \quad \text{(truncated at } \mathrm{D}^m\text{)}.$$

Example 3a

Solve $(\mathrm{D}^2 - 3\mathrm{D} + 2)y = x^2$.

Factor out the constant term 2:

$$y_{\mathrm{p}} = \frac{1}{2}\Big[1 - \big(\tfrac{3\mathrm{D}-\mathrm{D}^2}{2}\big)\Big]^{-1} x^2 = \frac{1}{2}\big(1 + u + u^2 + \dots\big)x^2, \qquad u = \tfrac{3\mathrm{D}}{2} - \tfrac{\mathrm{D}^2}{2}.$$

Keeping powers up to D^2: $1 + u + u^2 = 1 + \frac{3}{2}\mathrm{D} + \frac{7}{4}\mathrm{D}^2$. With $\mathrm{D}x^2 = 2x$, $\mathrm{D}^2x^2 = 2$,

$$y_{\mathrm{p}} = \frac{1}{2}\Big(x^2 + \tfrac{3}{2}\cdot 2x + \tfrac{7}{4}\cdot 2\Big) = \frac{x^2}{2} + \frac{3x}{2} + \frac{7}{4}.$$

Example 3b — lecture problem

Solve completely $y'' + 2y' + 2y = x^2$.

Operator form: $(\mathrm{D}^2 + 2\mathrm{D} + 2)y = x^2$.

Complementary function. $\mathrm{D}^2 + 2\mathrm{D} + 2 = 0$ has discriminant $4 - 8 = -4$, so $\mathrm{D} = -1 \pm i$ and

$$y_{\mathrm{c}} = \mathrm{e}^{-x}\big(C_1 \cos x + C_2 \sin x\big).$$

Particular integral. Make y the subject and factor out the constant term 2:

$$y_{\mathrm{p}} = \frac{1}{\mathrm{D}^2 + 2\mathrm{D} + 2}x^2 = \frac{1}{2}\Big[1 + \underbrace{\big(\tfrac{2\mathrm{D}+\mathrm{D}^2}{2}\big)}_{=u}\Big]^{-1} x^2 = \frac{1}{2}\big(1 - u + u^2 - \dots\big)x^2, \qquad u = \mathrm{D} + \tfrac{\mathrm{D}^2}{2}.$$

Since x^2 is degree 2, keep terms up to D^2. Now $u = \mathrm{D} + \frac{\mathrm{D}^2}{2}$ and $u^2 = \mathrm{D}^2 + (\text{higher})$, so on a quadratic

$$1 - u + u^2 = 1 - \mathrm{D} - \tfrac{\mathrm{D}^2}{2} + \mathrm{D}^2 = 1 - \mathrm{D} + \tfrac{1}{2}\mathrm{D}^2.$$

Apply to x^2 (with $\mathrm{D}x^2 = 2x$, $\mathrm{D}^2x^2 = 2$):

$$y_{\mathrm{p}} = \frac{1}{2}\Big(x^2 - 2x + \tfrac{1}{2}\cdot 2\Big) = \frac{1}{2}\big(x^2 - 2x + 1\big) = \frac{(x-1)^2}{2}.$$

Hence the general solution is

$$\boxed{y = \mathrm{e}^{-x}\big(C_1 \cos x + C_2 \sin x\big) + \tfrac{1}{2}\big(x^2 - 2x + 1\big).}$$

Check: $y_{\mathrm{p}}'' + 2y_{\mathrm{p}}' + 2y_{\mathrm{p}} = 1 + 2(x-1) + (x^2 - 2x + 1) = x^2$. ✓

Case 4 — Exponential times a function, $f(x) = \mathrm{e}^{ax}V(x)$

Pull the exponential out front and *shift* the operator's argument. This is the single most useful case — it converts "exponential × something" into a plain problem in V.

Case 4 (Shift theorem)

$$\frac{1}{\phi(\mathrm{D})}\left[\mathrm{e}^{ax}V(x)\right] = \mathrm{e}^{ax}\,\frac{1}{\phi(\mathrm{D}+a)}\,V(x).$$

Example 4

Solve $(\mathrm{D}^2+2\mathrm{D}+1)y = x\,\mathrm{e}^{-x}$.
Here $\phi(\mathrm{D}) = (\mathrm{D}+1)^2$ and $a=-1$. Shifting $\mathrm{D}\to\mathrm{D}-1$:

$$y_\mathrm{p} = \mathrm{e}^{-x}\frac{1}{(\mathrm{D}-1+1)^2}\,x = \mathrm{e}^{-x}\frac{1}{\mathrm{D}^2}\,x = \mathrm{e}^{-x}\iint x\,\mathrm{d}x\,\mathrm{d}x = \frac{x^3\mathrm{e}^{-x}}{6}.$$

The shift automatically resolves the double-root resonance (-1 is a repeated root), turning it into two clean integrations — compare with (**??**).

Case 5 — Input of the form $f(x) = x\,V(x)$

Case 5

$$\frac{1}{\phi(\mathrm{D})}\left[x\,V(x)\right] = \left[x - \frac{\phi'(\mathrm{D})}{\phi(\mathrm{D})}\right]\frac{1}{\phi(\mathrm{D})}\,V(x).$$

Example 5

Solve $(\mathrm{D}^2+4)y = x\sin x$, so $\phi(\mathrm{D}) = \mathrm{D}^2+4$ and $\phi'(\mathrm{D}) = 2\mathrm{D}$.
First $\dfrac{1}{\mathrm{D}^2+4}\sin x = \dfrac{1}{-1+4}\sin x = \frac{1}{3}\sin x$ (Case 2). Then

$$y_\mathrm{p} = \left[x - \frac{2\mathrm{D}}{\mathrm{D}^2+4}\right]\tfrac{1}{3}\sin x = \frac{x}{3}\sin x - \frac{2}{3}\cdot\frac{1}{\mathrm{D}^2+4}(\cos x) = \frac{x}{3}\sin x - \frac{2}{9}\cos x.$$

Case 6 — Anything else (partial fractions)

When $f(x)$ fits none of the templates above — $\tan x$, $\sec x$, $\dfrac{1}{1+\mathrm{e}^x}$ and the like — resolve $\dfrac{1}{\phi(\mathrm{D})}$ into **partial fractions in** D and apply the foundational formula $\dfrac{1}{\mathrm{D}-a}f = \mathrm{e}^{ax}\int \mathrm{e}^{-ax}f\,\mathrm{d}x$ to each piece.

Example 6

Solve $(\mathrm{D}^2+1)y = \sec x$.
Since $\mathrm{D}^2+1 = (\mathrm{D}-i)(\mathrm{D}+i)$, $\dfrac{1}{\mathrm{D}^2+1} = \dfrac{1}{2i}\left(\dfrac{1}{\mathrm{D}-i} - \dfrac{1}{\mathrm{D}+i}\right)$. Now

$$\frac{1}{\mathrm{D}-i}\sec x = \mathrm{e}^{ix}\int \mathrm{e}^{-ix}\sec x\,\mathrm{d}x = \mathrm{e}^{ix}\int (1 - i\tan x)\,\mathrm{d}x = \mathrm{e}^{ix}\big(x + i\ln|\cos x|\big),$$

and similarly for $\dfrac{1}{\mathrm{D}+i}\sec x$. Subtracting and dividing by $2i$,

$$y_{\mathrm{p}} = x\sin x + \cos x\,\ln|\cos x|,$$

the same result variation of parameters produces — reached here purely algebraically.

3. Summary of the toolkit

Quick reference — identify $f(x)$, then act

Right-hand side $f(x)$	What to do after making y the subject
e^{ax}	replace $\mathrm{D}\to a$: $\dfrac{1}{\phi(a)}\mathrm{e}^{ax}$; if $\phi(a)=0$, use $\dfrac{x^r}{r!\,\psi(a)}\mathrm{e}^{ax}$
$\sin ax,\ \cos ax$	replace $\mathrm{D}^2\to -a^2$; if it vanishes, $\mp\dfrac{x}{2a}\cos\,/\,\sin$
x^m	expand $[\phi(\mathrm{D})]^{-1}$ in powers of D, up to D^m
$\mathrm{e}^{ax}V(x)$	shift: $\mathrm{e}^{ax}\dfrac{1}{\phi(\mathrm{D}+a)}V$
$x\,V(x)$	$\Big[x-\phi'(\mathrm{D})/\phi(\mathrm{D})\Big]\dfrac{1}{\phi(\mathrm{D})}V$
anything else	partial fractions + $\dfrac{1}{\mathrm{D}-a}f=\mathrm{e}^{ax}\int\mathrm{e}^{-ax}f\,\mathrm{d}x$

Closing note. Because $\phi(\mathrm{D})$ and any $\dfrac{1}{\phi(\mathrm{D})}$ commute, the order in which you peel off factors is immaterial — though one order is often algebraically lighter. Always remember to add the complementary function: the full answer is $y = y_{\mathrm{c}} + y_{\mathrm{p}}$.

4. Graded exercise set

Attempt these in order; difficulty rises within each band. Full worked solutions follow overleaf — attempt every problem before consulting them.

Band A — Direct application of a single case

A1. Find the particular integral of $(\mathrm{D}^2-4)y=\mathrm{e}^{5x}$.

A2. Find the particular integral of $(\mathrm{D}^2+\mathrm{D}-6)y=\cos 3x$.

A3. Find the particular integral of $(\mathrm{D}^2+3\mathrm{D}+2)y=x^2+x$.

Band B — Resonance and the shift theorem

B1. Solve $(\mathrm{D}^2-4\mathrm{D}+4)y=\mathrm{e}^{2x}$ completely (i.e. give $y=y_{\mathrm{c}}+y_{\mathrm{p}}$).

B2. Find the particular integral of $(\mathrm{D}^2+9)y=\cos 3x$.

B3. Find the particular integral of $(\mathrm{D}^2-2\mathrm{D}+1)y=x^2\mathrm{e}^{x}$.

Band C — Products, general method, and synthesis

C1. Find the particular integral of $(\mathrm{D}^2+1)y = x\cos x$.

C2. Using the general method, find the particular integral of $(\mathrm{D}^2+1)y = \tan x$.

C3. Solve completely $(\mathrm{D}^2-1)y = \dfrac{2}{1+\mathrm{e}^x}$. *(Hint: partial fractions in* D*, then substitute* $u = \mathrm{e}^x$ *in the integrals.)*

5. Worked solutions

Band A

A1. $\phi(\mathrm{D}) = \mathrm{D}^2 - 4$, input e^{5x} (Case 1). $\phi(5) = 25 - 4 = 21 \neq 0$, so $y_\mathrm{p} = \dfrac{1}{21}\mathrm{e}^{5x}$.

A2. $\phi(\mathrm{D}) = \mathrm{D}^2 + \mathrm{D} - 6$, input $\cos 3x$ (Case 2). Replace $\mathrm{D}^2 \to -9$:

$$y_\mathrm{p} = \frac{1}{\mathrm{D} - 15}\cos 3x = \frac{\mathrm{D} + 15}{\mathrm{D}^2 - 225}\cos 3x = \frac{\mathrm{D} + 15}{-9 - 225}\cos 3x,$$

and with $\mathrm{D}\cos 3x = -3\sin 3x$,

$$y_\mathrm{p} = \frac{-3\sin 3x + 15\cos 3x}{-234} = \frac{\sin 3x - 5\cos 3x}{78}.$$

A3. $\phi(\mathrm{D}) = \mathrm{D}^2 + 3\mathrm{D} + 2$, input $x^2 + x$ (Case 3). Factor out 2: $\frac{1}{2}(1 - u + u^2 - \dots)$ with $u = \frac{3\mathrm{D}}{2} + \frac{\mathrm{D}^2}{2}$, giving up to D^2: $1 - \frac{3}{2}\mathrm{D} + \frac{7}{4}\mathrm{D}^2$. With $g = x^2 + x$, $\mathrm{D}g = 2x + 1$, $\mathrm{D}^2 g = 2$:

$$y_\mathrm{p} = \frac{1}{2}\Big[(x^2 + x) - \tfrac{3}{2}(2x + 1) + \tfrac{7}{4}(2)\Big] = \frac{x^2}{2} - x + 1.$$

Band B

B1. $\phi(\mathrm{D}) = (\mathrm{D} - 2)^2$; roots $2, 2$ give $y_\mathrm{c} = (c_1 + c_2 x)\mathrm{e}^{2x}$. The input e^{2x} hits a *double* root ($r = 2$), so by (**??**), $y_\mathrm{p} = \dfrac{x^2}{2!}\mathrm{e}^{2x} = \dfrac{x^2}{2}\mathrm{e}^{2x}$, and

$$y = (c_1 + c_2 x)\mathrm{e}^{2x} + \tfrac{x^2}{2}\mathrm{e}^{2x}.$$

B2. $\phi(\mathrm{D}) = \mathrm{D}^2 + 9$, input $\cos 3x$. Since $\phi(-9) = 0$ with $a = 3$ (resonant), $y_\mathrm{p} = \dfrac{x}{2 \cdot 3}\sin 3x = \dfrac{x}{6}\sin 3x$.

B3. $\phi(\mathrm{D}) = (\mathrm{D} - 1)^2$, input $x^2\mathrm{e}^x$ (Case 4, shift $a = 1$):

$$y_\mathrm{p} = \mathrm{e}^x \frac{1}{\mathrm{D}^2} x^2 = \mathrm{e}^x \iint x^2 \,\mathrm{d}x\,\mathrm{d}x = \frac{x^4 \mathrm{e}^x}{12}.$$

Band C

C1. $\phi(\mathrm{D}) = \mathrm{D}^2 + 1$, input $x\cos x$ (Case 5, $V = \cos x$, $\phi' = 2\mathrm{D}$). The complex route is cleanest: with $x\cos x = \mathrm{Re}\; x\mathrm{e}^{ix}$,

$$\frac{1}{\mathrm{D}^2 + 1} x\mathrm{e}^{ix} = \mathrm{e}^{ix}\frac{1}{(\mathrm{D} + i)^2 + 1} x = \mathrm{e}^{ix}\frac{1}{\mathrm{D}^2 + 2i\mathrm{D}} x = \mathrm{e}^{ix}\frac{1}{2i\mathrm{D}}\Big(1 - \tfrac{\mathrm{D}}{2i} + \dots\Big)x.$$

Then $\big(1 - \frac{\mathrm{D}}{2i}\big)x = x + \frac{i}{2}$, and $\dfrac{1}{2i\mathrm{D}}\big(x + \frac{i}{2}\big) = \dfrac{1}{2i}\big(\frac{x^2}{2} + \frac{i}{2}x\big) = -\frac{ix^2}{4} + \frac{x}{4}$. Multiplying by e^{ix} and taking the real part,

$$y_\mathrm{p} = \frac{x^2}{4}\sin x + \frac{x}{4}\cos x.$$

C2. $\phi(\mathrm{D}) = \mathrm{D}^2 + 1 = (\mathrm{D} - i)(\mathrm{D} + i)$, input $\tan x$ (Case 6). Resolving into partial fractions in D and evaluating the conjugate integrals (equivalently, variation of parameters) gives

$$y_\mathrm{p} = -\cos x \,\ln|\sec x + \tan x|.$$

(Students should confirm $y_{\mathrm{p}}'' + y_{\mathrm{p}} = \tan x$.)

C3. $\phi(\mathrm{D}) = \mathrm{D}^2 - 1 = (\mathrm{D}-1)(\mathrm{D}+1)$, input $g(x) = \dfrac{2}{1+\mathrm{e}^{x}}$. Partial fractions: $\dfrac{1}{\mathrm{D}^2-1} = \dfrac{1}{2}\Big(\dfrac{1}{\mathrm{D}-1} - \dfrac{1}{\mathrm{D}+1}\Big)$.

First piece. With $u = \mathrm{e}^{x}$,

$$\frac{1}{\mathrm{D}-1}g = \mathrm{e}^{x}\int \frac{2\mathrm{e}^{-x}}{1+\mathrm{e}^{x}}\,\mathrm{d}x = \mathrm{e}^{x}\int \frac{2\,\mathrm{d}u}{u^{2}(1+u)} = \mathrm{e}^{x}\big[-2\mathrm{e}^{-x} - 2x + 2\ln(1+\mathrm{e}^{x})\big].$$

Second piece. $\dfrac{1}{\mathrm{D}+1}g = \mathrm{e}^{-x}\displaystyle\int \frac{2\,\mathrm{d}u}{1+u} = 2\mathrm{e}^{-x}\ln(1+\mathrm{e}^{x})$.

Combine.

$$y_{\mathrm{p}} = \frac{1}{2}\Big[\big(-2 - 2x\mathrm{e}^{x} + 2\mathrm{e}^{x}\ln(1+\mathrm{e}^{x})\big) - 2\mathrm{e}^{-x}\ln(1+\mathrm{e}^{x})\Big] = -1 - x\mathrm{e}^{x} + \big(\mathrm{e}^{x} - \mathrm{e}^{-x}\big)\ln(1+\mathrm{e}^{x}).$$

Adding $y_{\mathrm{c}} = c_1\mathrm{e}^{x} + c_2\mathrm{e}^{-x}$ gives the general solution; the essential new content is the $\ln(1+\mathrm{e}^{x})$ combination.

End of handout — M3503, Department of Mathematics & Computer Science, NUL.